

A Rectangle Class with Magic Methods

To expand, we'll build a program that draws a number of rectangle shapes using a `Rectangle` class. The user will be able to click on any two rectangles, and the program will report if the rectangles have the same area or if the area of the first one is larger or smaller than the area of the second rectangle. We'll use the `==`, `<`, and `>` operators and expect the result to be a Boolean `True` or `False` for each comparison. [Listing 9-6](#) contains the code of the `Rectangle` class, which implements magic methods for these operators.

File: MagicMethods/Rectangle/Rectangle.py

```
# Rectangle class

import pygame
import random

# Set up the colors
RED = (255, 0, 0)
GREEN = (0, 255, 0)
BLUE = (0, 0, 255)

class Rectangle():

    def __init__(self, window):
        self.window = window
        self.width = random.choice((20, 30, 40))
        self.height = random.choice((20, 30, 40))
        self.color = random.choice((RED, GREEN, BLUE))
        self.x = random.randrange(0, 400)
        self.y = random.randrange(0, 400)
        self.rect = pygame.Rect(self.x, self.y, self.width, self.height)
        self.area = self.width * self.height

    def clickedInside(self, mousePoint):
        clicked = self.rect.collidepoint(mousePoint)
        return clicked

# Magic method called when you compare
# two Rectangle objects with the == operator
def __eq__(self, oOtherRectangle): ❶
    if not isinstance(oOtherRectangle, Rectangle):
```